

KENTARO TAKAHIRA

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Ph.D. candidate at Hong Kong University of Science and Technology, specializing in Data Visualization and AR/VR, with four years of industry experience as a data scientist. Author of a book on data visualization.

EDUCATION

Hong Kong University of Science and Technology

Hong Kong

PhD in Computer Science and Engineering

09/2022 – present

- ◇ Research Areas: Data Visualization
- ◇ Supervisor: Huamin Qu

Kyoto University

Kyoto, Japan

MSc in Informatics

04/2016 – 03/2017 & 04/2018 – 03/2019

- ◇ Research Areas: Artificial Intelligence
- ◇ Supervisor: Shigeo Matsubara

Kyoto University

Kyoto, Japan

BSc in Economics

04/2012 – 03/2016

- ◇ Specialization: Statistics and Econometrics
- ◇ Supervisor: Yoshihiko Nishiyama

KTH Royal Institute of Technology

Stockholm, Sweden

Exchange Program in School of Engineering Sciences

09/2014 – 06/2015

WORKING EXPERIENCE

Freelance

Tokyo, Japan

Data Analyst

11/2021 – present

- ◇ Analyzed client data in the retail and manufacturing sectors.
- ◇ Delivered corporate training sessions and lectures on data analysis and visualization.

Veldt Inc.

Tokyo, Japan

Data Scientist

04/2021 – 10/2021

- ◇ Conducted R&D in edge computing and causal inference techniques.
- ◇ Designed and implemented data pipelines and analytics frameworks for wearable IoT devices.

Accenture

Tokyo, Japan

Data Scientist

08/2019 – 03/2021

- ◇ Developed machine learning solutions and statistical models for enterprise-level applications.
- ◇ Led end-to-end pipeline development including data preprocessing, model training, and deployment.

Actavo Inc.

Wakefield, UK

Data Analyst Trainee

08/2017 – 03/2018

- ◇ Performed statistical modeling and data management for field operations.
- ◇ Traineeship supported by the EU-Japan Centre for Industrial Cooperation.

NTT Basic Research Laboratories

Tokyo, Japan

Research Scientist Intern

08/2016 – 09/2016

- ◇ Analyzed urban sensor data as part of the Tokyo Smart City Project.
- ◇ Investigated infrastructure and mobility patterns using real-world datasets.

CONFERENCE PUBLICATIONS

Takahira, K., Yu, Y., Fujiwara, T., Suzuki, R., Qu, H. **InSituTale: Enhancing Augmented Data Storytelling with Physical Objects**. In Proceedings of the 37th Annual ACM Symposium on User Interface Software and Technology. ACM, 2025. (**UIST '25**, acceptance rate: 22%)

Takahira, K., Kam-Kwai, W., Yang, L., Xu, X., Fujiwara, T., Qu, H. **TangibleNet: Synchronous Network Data Storytelling through Tangible Interactions in Augmented Reality**. In Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems. ACM, 2025. (**CHI '25**, acceptance rate: 24%)

Shigyo, L., Cao, Y.F., **Takahira, K.**, Fan, M., Qu, H. **VR-Mediated Cognitive Defusion: A Comparative Study for Managing Negative Thoughts**. In Proceedings of the 32nd ACM International Conference on Multimedia, pp. 117–126. 2024. ACM, 2025. (**MM '24**, acceptance rate: 25%) **Honorable Mention Paper Award** (top 1%)

Takahira, K., Matsubara, S. **Contract-Based Inter-User Usage Coordination in Free-Floating Car Sharing**. In Proceedings of the AAAI Conference on Artificial Intelligence, 2021. (**AAAI '21**, acceptance rate: 21%)

JOURNAL PUBLICATIONS

Oishi, A., Teshima, T., Kojima, N., Kiriha, M., Kojima, N., Sasaki, T., **Takahira, K.**, Takeuchi, T., Tajima, K., Noda, C., Hirose, H., Yamanaka, S. **Forecasting the Movement of Internally Displaced People by Artificial Intelligence**. In *Digital Innovations, Business, and Society in Africa: New Frontiers and a Shared Strategic Vision*, Springer, 2022.

WORKSHOP AND POSTER PUBLICATIONS

Yuan, L., Tong, W., **Takahira, K.**, Wen, Z., Yang, Y., Qu, H. **Echoes in the Gallery: A Collaborative Immersive Analytics System for Analyzing Audience Reactions in Virtual Reality Exhibitions**. Workshop paper presented at IEEE Visualization Conference (VIS), 2023.

Li, H., **Takahira, K.**, Wong, K. K., Yang, L., Tong, W., Qu, H. **Landslide Visualization Situated on Tangible Terrain Models**. Poster presented at IEEE Visualization Conference (VIS), 2023.

Takahira, K., Matsubara, S. **Self-organized Auction-based Regulation in Free-Floating Car Sharing**. Poster presented at Joint Agent Workshops and Symposium (JAWS), co-organized with IEEE Computer Society Japan Chapter., 2018.

INVITED TALKS

- ◇ Osaka Seikei University, Osaka, Japan (hosted by Hiroaki Natsukawa), *May. 2025*
- ◇ Kyoto University, Kyoto, Japan (hosted by Yoshihiko Nishiyama), *Mar. 2016*

ACADEMIC AWARDS

- ◇ ACM Multimedia Honorable Mention Award *2024*
- ◇ Hong Kong PhD Fellowship Scheme *2022*
- ◇ Excellent Dissertation Award *Kyoto University 2019*
- ◇ Vulcanus in Europe Scholarship *2017*
- ◇ Keidanren Global Scholarship *2014*

COURSES AND BOOKS FOR GENERAL AUDIENCES

- ◇ Book: Data Visualization in Practice, Gijutsu-Hyohron Co., Ltd., *2025*
- ◇ Online Course: Practical Graph Design, Schoo, *2022*
- ◇ Online Course: Graph Design with Python, Schoo, *2022*